

Title: Multi-Objective Reinforcement Learning for Computing all Optimal Policies with Many Objectives

Tutor: Manel Rodríguez Soto

Abstract:

The purpose of this thesis is to investigate the formal and algorithmic frontiers of multi-objective reinforcement learning (MORL), an area with applications ranging from Large Language Models to video games and value alignment. MORL is currently considered the best framework for dealing with multi-objective sequential decision making problems. In more detail, the main objective of this work is to develop novel MORL algorithms capable of simultaneously computing multiple policies (i.e., behaviours) that take into account more than two objectives. To achieve this, the research is structured along two complementary fronts. First, we identify and analyse complex reinforcement learning environments that involve multiple objectives. Second, we design and implement a novel algorithm specifically crafted to handle prioritisation across many objectives: Lexicographic Hull Value Iteration (LHVI). This novel algorithms exploit state-of-the-art formal research on multi-objective reinforcement learning. Thereafter, we provide a detailed evaluation of LHVI against other state-of-the-art MORL algorithms such as Convex Hull Value Iteration (CHVI). The empirical results of this research will provide a baseline for future research in many-objective reinforcement learning.